

VESSEL ANALYSIS REPORT

vaf(%)	branch points count	tip point count	vein count	total vein length	average vein length
1.478	78.0	202.0	215.0	4215.627	19.608

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
1	[45,3]	[3,4]	42.243	BRANCH	BRANCH
2	[45,3]	[46,7]	3.414	BRANCH	BRANCH
3	[45,3]	[51,4]	5.828	BRANCH	BRANCH
4	[0,11]	[3,4]	7.243	TIP	BRANCH
5	[51,4]	[83,9]	33.899	BRANCH	BRANCH
6	[51,4]	[46,7]	5.828	BRANCH	BRANCH
7	[170,0]	[198,4]	28.657	TIP	BRANCH
8	[226,15]	[198,4]	34.142	TIP	BRANCH
9	[202,3]	[198,4]	3.0	TIP	BRANCH
10	[46,7]	[28,17]	21.728	BRANCH	BRANCH
11	[87,0]	[83,9]	9.657	TIP	BRANCH
12	[83,9]	[75,39]	32.314	BRANCH	BRANCH
13	[242,10]	[248,9]	5.414	TIP	BRANCH
14	[239,45]	[248,9]	38.728	TIP	BRANCH
15	[250,7]	[248,9]	1.414	TIP	BRANCH
16	[28,17]	[4,36]	37.556	BRANCH	BRANCH
17	[28,17]	[29,30]	12.414	BRANCH	BRANCH
18	[328,0]	[309,20]	27.456	TIP	BRANCH
19	[298,27]	[309,20]	12.899	TIP	BRANCH
20	[314,22]	[309,20]	4.414	TIP	BRANCH
21	[132,21]	[134,21]	1.0	BRANCH	BRANCH
22	[127,19]	[132,21]	4.828	TIP	BRANCH
23	[132,24]	[132,21]	2.0	TIP	BRANCH
24	[135,16]	[134,21]	4.414	TIP	BRANCH
25	[142,29]	[134,21]	12.071	TIP	BRANCH
26	[29,30]	[34,30]	4.0	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
27	[26,33]	[29,30]	2.828	TIP	BRANCH
28	[34,30]	[37,39]	9.243	BRANCH	BRANCH
29	[50,25]	[34,30]	16.657	TIP	BRANCH
30	[104,27]	[103,30]	2.414	TIP	BRANCH
31	[100,34]	[103,30]	3.828	TIP	BRANCH
32	[107,34]	[103,30]	4.243	TIP	BRANCH
33	[9,36]	[4,36]	4.828	TIP	BRANCH
34	[135,61]	[161,37]	36.698	TIP	BRANCH
35	[160,32]	[161,37]	4.0	TIP	BRANCH
36	[167,33]	[161,37]	6.243	TIP	BRANCH
37	[37,39]	[48,47]	14.485	BRANCH	BRANCH
38	[26,50]	[37,39]	23.243	TIP	BRANCH
39	[62,40]	[75,39]	12.414	TIP	BRANCH
40	[75,39]	[65,50]	17.071	BRANCH	BRANCH
41	[399,34]	[441,40]	44.314	TIP	BRANCH
42	[440,61]	[441,40]	20.414	TIP	BRANCH
43	[485,0]	[441,40]	60.326	TIP	BRANCH
44	[48,47]	[65,50]	17.243	BRANCH	BRANCH
45	[48,47]	[39,84]	40.142	BRANCH	BRANCH
46	[65,50]	[65,61]	10.0	BRANCH	BRANCH
47	[205,48]	[206,57]	9.243	TIP	BRANCH
48	[223,63]	[206,57]	18.485	TIP	BRANCH
49	[202,57]	[206,57]	3.0	TIP	BRANCH
50	[57,73]	[65,61]	15.485	TIP	BRANCH
51	[65,61]	[70,76]	15.657	BRANCH	BRANCH
52	[77,75]	[69,76]	7.414	TIP	BRANCH
53	[65,77]	[69,76]	3.414	TIP	BRANCH
54	[69,76]	[45,93]	32.385	BRANCH	BRANCH
55	[108,82]	[111,82]	2.0	TIP	BRANCH
56	[110,50]	[111,82]	75.77	TIP	BRANCH
57	[39,84]	[40,92]	7.414	BRANCH	BRANCH
58	[42,79]	[39,84]	4.828	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
59	[35,92]	[40,92]	4.0	TIP	BRANCH
60	[40,92]	[45,93]	4.0	BRANCH	BRANCH
61	[45,93]	[46,106]	12.414	BRANCH	BRANCH
62	[1,99]	[11,99]	9.828	TIP	BRANCH
63	[11,99]	[11,104]	4.0	BRANCH	BRANCH
64	[0,58]	[11,99]	52.87	TIP	BRANCH
65	[11,104]	[4,109]	8.657	BRANCH	BRANCH
66	[25,111]	[11,104]	15.485	TIP	BRANCH
67	[98,41]	[46,106]	102.669	TIP	BRANCH
68	[46,106]	[32,122]	21.556	BRANCH	BRANCH
69	[418,94]	[353,107]	72.698	TIP	BRANCH
70	[353,107]	[349,112]	5.657	BRANCH	BRANCH
71	[352,105]	[353,107]	1.0	TIP	BRANCH
72	[1,116]	[4,109]	7.243	TIP	BRANCH
73	[0,104]	[4,109]	5.243	TIP	BRANCH
74	[301,109]	[308,111]	6.828	TIP	BRANCH
75	[308,111]	[317,118]	12.071	BRANCH	BRANCH
76	[351,58]	[308,111]	70.569	TIP	BRANCH
77	[349,112]	[317,118]	33.485	BRANCH	BRANCH
78	[349,117]	[349,112]	4.0	TIP	BRANCH
79	[139,108]	[136,116]	8.243	TIP	BRANCH
80	[132,116]	[136,116]	3.0	TIP	BRANCH
81	[136,116]	[153,143]	35.799	BRANCH	BRANCH
82	[320,137]	[317,118]	20.657	TIP	BRANCH
83	[96,117]	[91,120]	5.243	TIP	BRANCH
84	[90,123]	[91,120]	2.414	TIP	BRANCH
85	[89,101]	[91,120]	18.414	TIP	BRANCH
86	[462,116]	[463,121]	4.414	TIP	BRANCH
87	[511,117]	[463,121]	50.314	TIP	BRANCH
88	[447,133]	[463,121]	20.556	TIP	BRANCH
89	[29,118]	[32,122]	4.828	TIP	BRANCH
90	[32,122]	[53,145]	33.042	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
91	[58,122]	[53,123]	4.414	TIP	BRANCH
92	[49,122]	[53,123]	3.414	TIP	BRANCH
93	[53,123]	[53,145]	22.657	BRANCH	BRANCH
94	[146,141]	[153,143]	6.828	TIP	BRANCH
95	[153,143]	[192,143]	40.899	BRANCH	BRANCH
96	[208,135]	[192,143]	18.899	TIP	BRANCH
97	[192,143]	[187,158]	16.071	BRANCH	BRANCH
98	[53,145]	[55,149]	3.828	BRANCH	BRANCH
99	[103,145]	[92,147]	10.828	TIP	BRANCH
100	[86,129]	[92,147]	22.657	TIP	BRANCH
101	[92,147]	[110,150]	24.728	BRANCH	BRANCH
102	[67,147]	[55,149]	12.657	TIP	BRANCH
103	[55,149]	[55,152]	2.0	BRANCH	BRANCH
104	[112,144]	[110,150]	7.828	TIP	BRANCH
105	[108,148]	[110,150]	1.414	TIP	BRANCH
106	[55,152]	[68,179]	33.142	BRANCH	BRANCH
107	[55,152]	[40,162]	18.142	BRANCH	BRANCH
108	[151,181]	[187,158]	44.527	TIP	BRANCH
109	[190,159]	[187,158]	2.0	TIP	BRANCH
110	[1,138]	[40,162]	50.284	TIP	BRANCH
111	[40,162]	[45,173]	11.657	BRANCH	BRANCH
112	[252,158]	[253,163]	4.414	TIP	BRANCH
113	[253,163]	[267,166]	14.243	BRANCH	BRANCH
114	[219,177]	[253,163]	38.385	TIP	BRANCH
115	[300,127]	[267,166]	54.598	TIP	BRANCH
116	[266,178]	[267,166]	11.414	TIP	BRANCH
117	[45,173]	[67,183]	25.142	BRANCH	BRANCH
118	[42,183]	[45,173]	10.828	TIP	BRANCH
119	[87,181]	[68,179]	18.828	TIP	BRANCH
120	[68,179]	[67,183]	3.414	BRANCH	BRANCH
121	[67,183]	[93,194]	29.728	BRANCH	BRANCH
122	[149,186]	[149,190]	3.0	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
123	[152,190]	[149,190]	2.0	TIP	BRANCH
124	[149,190]	[145,193]	3.828	BRANCH	BRANCH
125	[127,190]	[145,193]	18.243	TIP	BRANCH
126	[127,252]	[145,193]	69.355	TIP	BRANCH
127	[101,185]	[93,194]	11.485	TIP	BRANCH
128	[120,219]	[93,194]	37.113	TIP	BRANCH
129	[32,202]	[34,202]	13.485	BRANCH	PATH
130	[32,202]	[13,220]	26.042	BRANCH	BRANCH
131	[0,228]	[13,220]	15.314	TIP	BRANCH
132	[13,220]	[47,227]	37.142	BRANCH	BRANCH
133	[62,207]	[47,227]	25.799	TIP	BRANCH
134	[47,227]	[47,241]	13.0	BRANCH	BRANCH
135	[21,240]	[5,249]	25.657	BRANCH	BRANCH
136	[21,240]	[31,240]	9.0	BRANCH	BRANCH
137	[21,240]	[7,249]	17.899	BRANCH	BRANCH
138	[34,250]	[31,240]	10.243	TIP	BRANCH
139	[38,238]	[31,240]	6.414	TIP	BRANCH
140	[43,241]	[47,241]	3.0	TIP	BRANCH
141	[49,257]	[47,241]	15.828	TIP	BRANCH
142	[5,249]	[7,249]	1.0	BRANCH	BRANCH
143	[4,278]	[5,249]	29.828	TIP	BRANCH
144	[7,249]	[19,265]	20.556	BRANCH	BRANCH
145	[23,264]	[19,265]	3.414	TIP	BRANCH
146	[19,265]	[19,268]	2.0	BRANCH	BRANCH
147	[10,273]	[19,268]	10.071	TIP	BRANCH
148	[28,330]	[19,268]	69.284	TIP	BRANCH
149	[272,305]	[274,313]	7.828	TIP	BRANCH
150	[270,314]	[274,313]	3.414	TIP	BRANCH
151	[283,315]	[274,313]	8.414	TIP	BRANCH
152	[188,369]	[188,374]	4.828	TIP	BRANCH
153	[183,371]	[188,374]	5.243	TIP	BRANCH
154	[193,375]	[188,374]	4.0	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
155	[511,395]	[507,404]	9.243	TIP	BRANCH
156	[507,404]	[454,432]	63.184	BRANCH	BRANCH
157	[509,405]	[507,404]	1.0	TIP	BRANCH
158	[161,403]	[143,410]	19.899	TIP	BRANCH
159	[136,422]	[143,410]	13.899	TIP	BRANCH
160	[45,425]	[80,419]	36.485	TIP	BRANCH
161	[81,417]	[80,419]	1.0	TIP	BRANCH
162	[84,422]	[80,419]	3.828	TIP	BRANCH
163	[450,411]	[454,432]	21.657	TIP	BRANCH
164	[454,432]	[433,452]	30.042	BRANCH	BRANCH
165	[353,426]	[353,449]	23.657	TIP	BRANCH
166	[406,470]	[353,449]	60.698	TIP	BRANCH
167	[353,449]	[325,464]	32.799	BRANCH	BRANCH
168	[422,492]	[433,452]	44.385	TIP	BRANCH
169	[417,448]	[433,452]	16.243	TIP	BRANCH
170	[325,464]	[301,467]	24.243	BRANCH	BRANCH
171	[326,468]	[325,464]	3.414	TIP	BRANCH
172	[293,438]	[301,467]	31.314	TIP	BRANCH
173	[300,469]	[301,467]	1.414	TIP	BRANCH
174	[117,0]	[104,2]	13.828	TIP	TIP
175	[278,0]	[285,38]	40.899	TIP	TIP
176	[333,0]	[341,3]	9.243	TIP	TIP
177	[382,0]	[367,19]	25.799	TIP	TIP
178	[402,0]	[407,16]	18.071	TIP	TIP
179	[356,34]	[331,56]	37.042	TIP	TIP
180	[380,40]	[370,51]	16.899	TIP	TIP
181	[286,42]	[313,73]	44.527	TIP	TIP
182	[282,45]	[262,48]	22.071	TIP	TIP
183	[185,56]	[170,71]	30.485	TIP	TIP
184	[511,57]	[495,66]	19.728	TIP	TIP
185	[186,81]	[172,84]	19.485	TIP	TIP
186	[256,81]	[280,108]	44.799	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
187	[141,90]	[131,98]	13.899	TIP	TIP
188	[153,92]	[154,103]	17.071	TIP	TIP
189	[171,98]	[195,110]	40.142	TIP	TIP
190	[110,125]	[142,137]	38.627	TIP	TIP
191	[417,146]	[397,164]	28.627	TIP	TIP
192	[15,173]	[26,180]	13.899	TIP	TIP
193	[15,185]	[17,195]	10.828	TIP	TIP
194	[196,188]	[173,201]	28.971	TIP	TIP
195	[301,201]	[301,222]	21.828	TIP	TIP
196	[217,210]	[160,231]	66.527	TIP	TIP
197	[307,245]	[354,283]	74.113	TIP	TIP
198	[511,246]	[503,260]	17.314	TIP	TIP
199	[77,266]	[87,281]	19.728	TIP	TIP
200	[19,303]	[15,315]	13.657	TIP	TIP
201	[413,318]	[435,319]	23.243	TIP	TIP
202	[380,326]	[345,347]	43.698	TIP	TIP
203	[226,344]	[217,348]	10.657	TIP	TIP
204	[16,356]	[10,376]	22.485	TIP	TIP
205	[262,362]	[277,378]	22.799	TIP	TIP
206	[43,366]	[62,382]	30.556	TIP	TIP
207	[102,366]	[95,376]	12.899	TIP	TIP
208	[9,380]	[0,409]	32.728	TIP	TIP
209	[254,406]	[252,415]	9.828	TIP	TIP
210	[105,423]	[106,433]	12.071	TIP	TIP
211	[117,439]	[145,452]	34.213	TIP	TIP
212	[56,457]	[57,472]	18.728	TIP	TIP
213	[153,479]	[159,489]	13.314	TIP	TIP
214	[291,482]	[280,490]	14.899	TIP	TIP
215	[81,504]	[98,511]	19.899	TIP	TIP